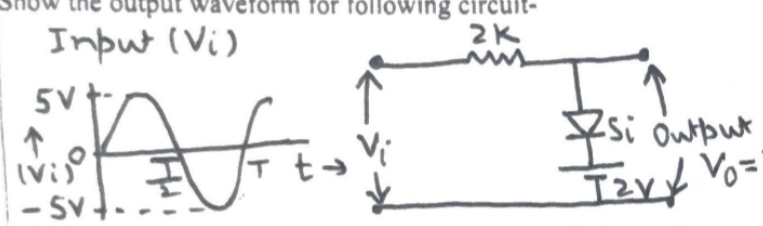
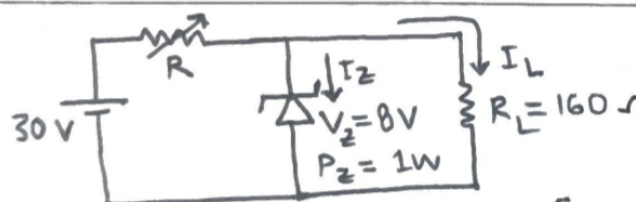


UNIT 2: Transistors, FETs and Operational Amplifiers

Syllabus topics: Basic theory and operation of PNP and NPN transistors, Characteristics of Common Base, Common Emitter and Common Collector configuration, DC Biasing: Fixed Bias, Emitter Bias, voltage divider bias, Field effect transistor: JFET, Drain and Transfer characteristic, MOSFET, Introduction to Operational Amplifier and its Applications as Adder, Subtractor, Integrator, Differentiator, log antilog.

Question 1

1.	Explain the mechanisms to explain the breakdown mechanisms in diode characteristic.	4	1	K2
2.	Establish relation between α and β . Evaluate change in α , if β is increased from 51 to 80. Or Show the output waveform for following circuit- Input (V_i) 	3	1,2	K3
3.	Draw and explain input and output characteristic curves for the Common Base Configuration. Or Explain the working of Bridge Rectifier. Also evaluate efficiency and the PIV.	4	1,2	K2
4.		4	1	K5

MID TERM-II | NET 101 | 2025-26 | Q2 Relation between α and β / evaluate change in α when β goes 51 $\rightarrow$ 80 OR output waveform for clipper circuit, Q3 Input/output characteristics of Common Base configuration OR Bridge rectifier efficiency & PIV

Question 2

	voltage and variable load.	3	CO1
4	Explain working of BJT in CB configuration and sketch the input and output characteristics. Also indicate active region, cutoff region and saturation region on the output characteristic curve.	3	CO2
5	Find I_C and V_{CE} for the circuit of Figure Q. 5, assuming $\beta=100$. Assuming $I_C \cong I_E$.	3	CO3

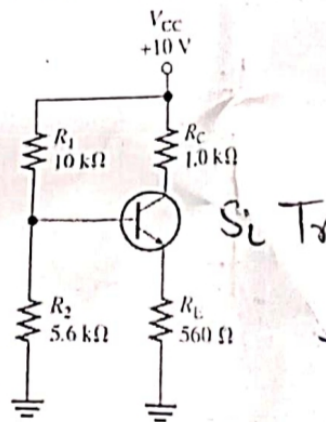


Figure Q. 5

Ask Gemini



MID SEM-2 | NET-101 | CE/CSE/EE/ET/IT/ME | Q4 BJT in CB configuration — input/output characteristics, active/cutoff/saturation regions, Q5 Find I_C and V_{CE} for given circuit ($\beta=100$, assuming $I_C \cong I_E$)

Question 3

1 For the network given in Figure 1, determine (i) I_B , (ii) I_C , (iii) β , and (iv) V_{CE}

3

CO2 K1

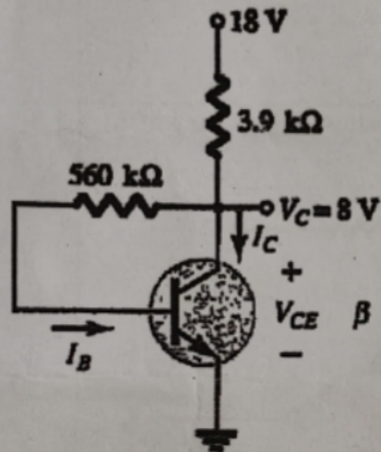


Fig. 1

2 For the circuit given below in Figure 2, determine- (i) I_{BQ} , (ii) I_{CQ} , (iii) V_{CEQ} and (iv) V_E .

3

CO2 K4, 5

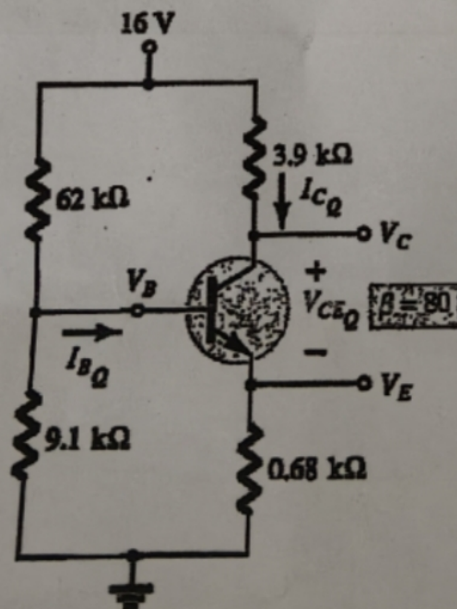


Fig. 2

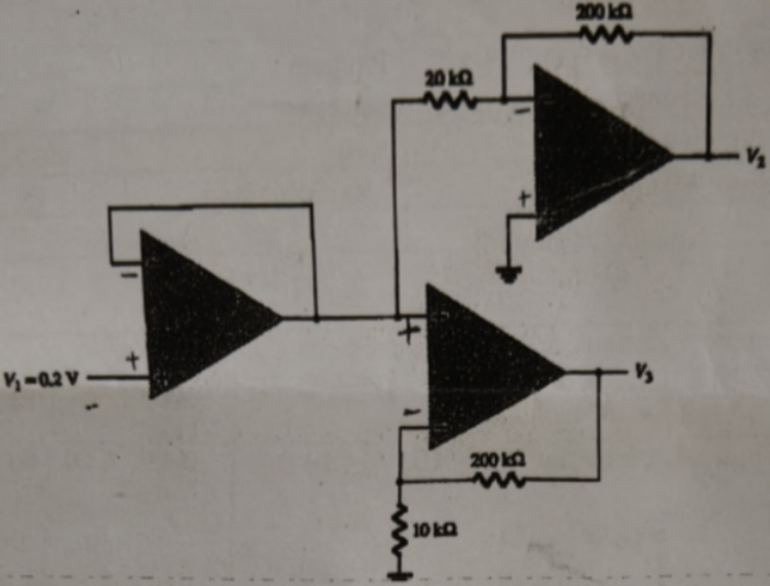
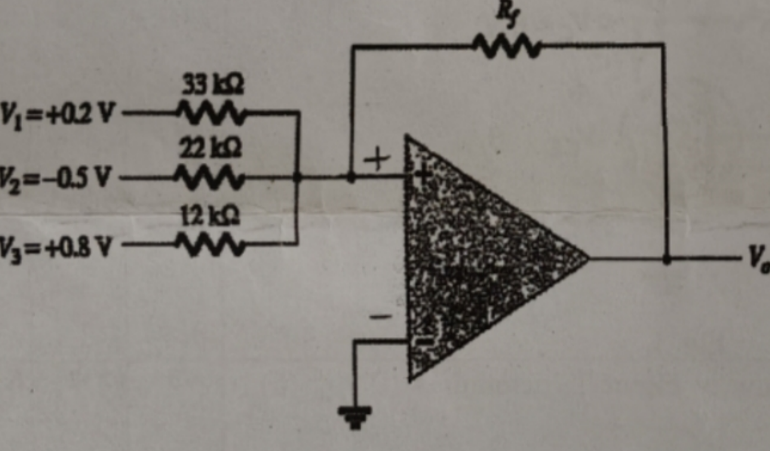
3 Explain the working of N- Channel Enhancement type MOSFET and also draw its transfer and drain characteristics.

3

CO2 K3, 5

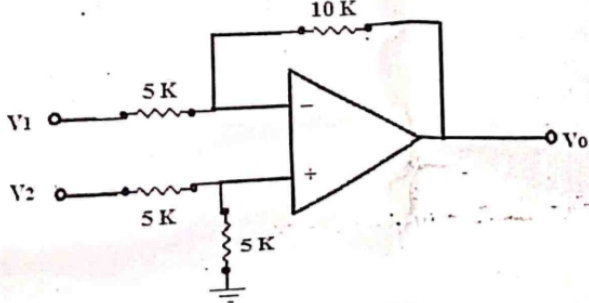
MID TERM-II | NET-101 | CE/CSE/EE/ET/IT/ME | Q1 Find I_B , I_C , β , V_{CE} for given network (Fig.1), Q2 Find I_{BQ} , I_{CQ} , V_{CEQ} , V_E for given circuit (Fig.2), Q3 N-channel enhancement MOSFET working + transfer & drain characteristics

Question 4

4	Determine V_2 and V_3 in below circuit shown in Figure 3.	3	CO2	K1, 5
 Fig.3		3	CO2	K4, 5
5	Calculate the output voltage (V_o) for below circuit shown in Figure 4, given that $R_f = 330K\Omega$.	3	CO2	K4, 5
 Fig.4				

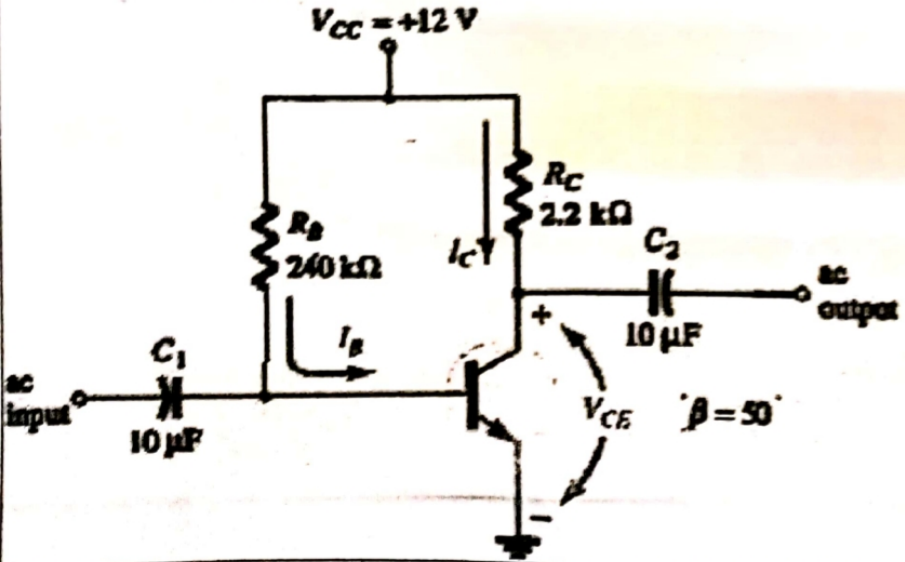
MID TERM-II | NET-101 (same paper, continued) | Q4 Determine V_2 and V_3 for cascaded op-amp circuit (Fig.3), Q5 Calculate output voltage V_o for summing op-amp circuit (Fig.4), $R_f = 330 k\Omega$

Question 5

2(a)	Explain the construction and working operation of N Channel Enhancement type MOSFET with suitable diagram. Also draw its drain and transfer characteristics.	5	CO2	K1, K2
2(b)	What are the properties of an ideal op-amp. Find the output voltage v_o for the Op-Amp circuit given below, where $v_1=10$ volt and $v_2=5$ volt.	5	CO2	K4, K5
				
3(a)	Find the minimal expression and realize it with basic gates. $F(A,B,C,D)=\sum m(0,1,3,5,7,9,10,14,15) + d(2,4)$	5	CO3	K5, K6
3(b)	Explain S-R flip flop with suitable diagram and show the Truth Table, Characteristic Table, Characteristic Equation and Excitation Table.	5	CO3	K3, K4

END SEM EXAM 2024-25 | NET-101 | CE/CSE/EE/ET/IT/ME | Q2(a) N-Channel Enhancement MOSFET construction & working + drain/transfer characteristics, Q2(b) Properties of ideal op-amp; find output voltage V_o

Question 6

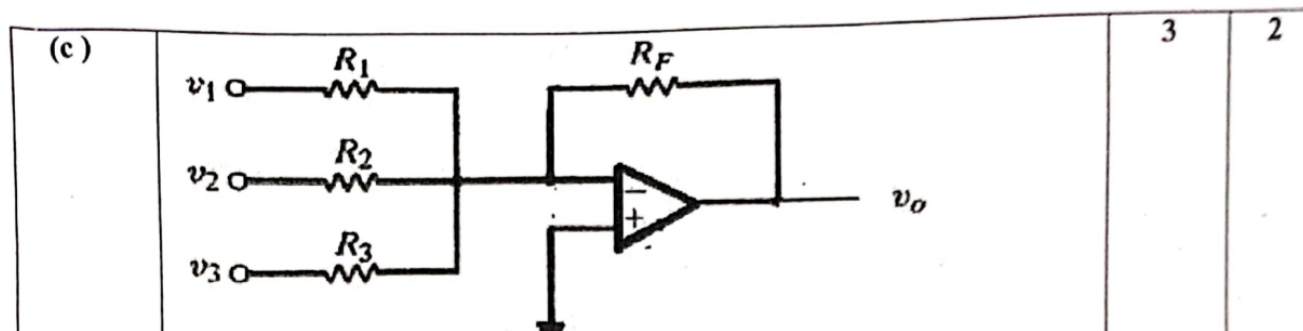
2(a)	Explain the input and output characteristic of Common Emitter (CE) amplifier with supporting diagrams. Write the relation between α and β .	4	2
(b)	Calculate the I_{BQ} , I_{CQ} and V_{CEQ} for following circuit-	3	2
			

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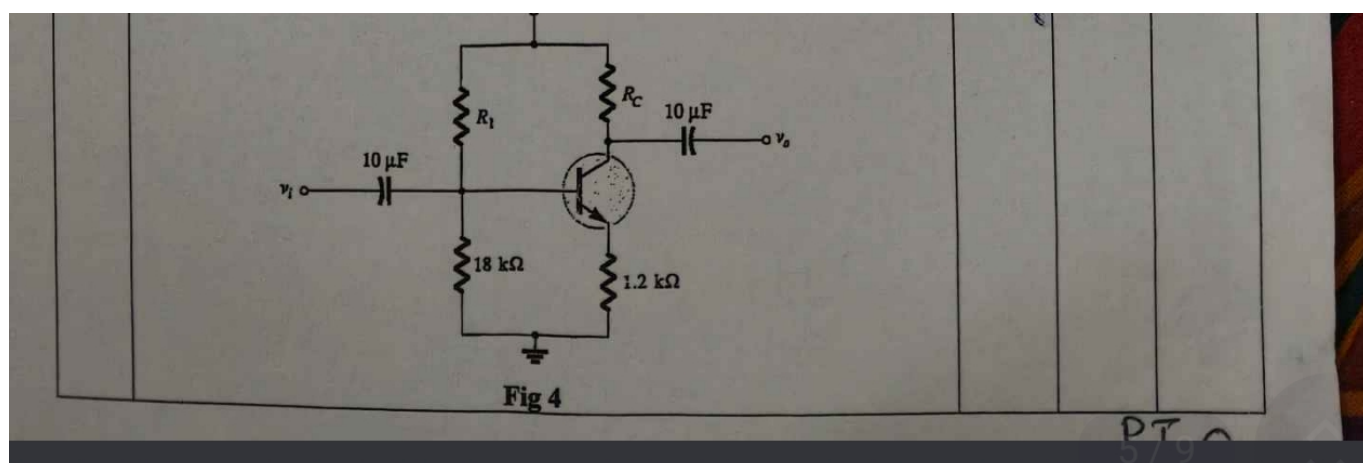
END SEM PYQ's | NET101 | CS/CE/EC/EE/IT/ME | Q2(a) CE amplifier input/output characteristics + relation between α and β , Q2(b) Calculate I_{BQ} , I_{CQ} , V_{CEQ} for given circuit

Question 7



END SEM PYQ's | NET101 (same paper, continued) | Q2(c) Summing op-amp circuit — find V_o for $v_1=x$, $v_2=2x$, $v_3=3x$

Question 8



Practice / Model Question Set | NET-101 | Q2(a) BJT bias circuit — find R_1 & R_C given $I_C=2\text{mA}$, $V_{CE}=10\text{V}$, $\beta=200$

Question 9

Q2(b)	Draw the construction and working of N-Channel JFET. Draw its transfer and characteristics graph.	5	CO2	K1
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Practice / Model Question Set (same paper, continued) | Q2(b) N-Channel JFET construction & working + transfer & characteristics graph